

5 Stretch Training-Related Range of Motion (ROM) Changes and Mechanisms

Chronic stretching (training) can increase range of motion (ROM) using different stretching techniques, positions, and durations (9). Naturally, there are always dissenting findings as for example; four (10 repetitions of 30 seconds; 3 days/week) (19) or six (four repetitions of 45 seconds; four days/week) (2) weeks of hamstrings static stretching did not increase hip extension or flexion ROM respectively. But the vast majority of stretch training programmes do exhibit ROM improvements (4, 7, 23, 26, 32). The most recent meta-analysis on the effects of chronic stretch training on ROM by the “king” of stretching and foam rolling meta-analyses, my friend and colleague Andreas Konrad reported that based on 77 studies that stretch training can provide moderate magnitude, effect size ROM increases (16).

But which type of stretch training is most effective? A review by Decoster et al. (9) indicated that static stretch training provided greater ROM improvements than proprioceptive neuromuscular facilitation (PNF). Other studies have documented static stretch-induced ROM improvements with no improvements with PNF (30 second stretch; three days/week for four weeks) (7), whereas another study found training-induced ROM improvements with no difference between static and PNF stretching (four days/week for six weeks) (32). While one study reported more than double the ROM improvements with static stretching versus dynamic stretching (1), others have not shown any significant difference (31). Once again, it is important to quantify the literature in general (meta-analysis) as reading individual studies can be quite confusing. Going back to the Konrad et al.'s meta-analysis (16), their subgroup analyses showed no difference in effectiveness between chronic static and PNF stretch training programmes, but both of these stretches produced greater ROM than ballistic and dynamic stretching. Furthermore, neither stretching volume, intensity, or weekly frequency demonstrated significant differences in ROM gains.

Their final significant finding was the evidence for greater training-related improvements in female ROM compared to males (16). This sex difference for stretch training contrasts with another meta-analysis (47 studies) of changes in ROM with an acute (single) bout of stretching, which did not find a sex difference (3). Thus, while women tend to possess greater absolute or inherent

flexibility than men and when performing chronic stretch training can achieve relatively greater ROM gains, they do not show such superiority with the relative flexibility increases following a single, acute session of stretching. In our acute meta-analysis (3), we suggested that although baseline flexibility is more limited in young or older adult males compared to females, the potential for acute ROM increases is not hindered by age, sex, or trained state. For the chronic stretch training results where women experienced greater relative ROM gains, we speculated that females do not exhibit higher ankle joint baseline flexibility compared to males. As the ankle joint was the most frequently tested joint in the literature, then the males would not have this explanation of a lower baseline from which to achieve higher relative increases. It was also postulated that women and girls may be more sensitive to chronic stretch training than men and boys.

The persistence of these flexibility adaptations after stretch-training desists have been reported to show better than pre-training ROM for three (20), four (29, 33), and eight (24) weeks. Similar to strength training, a maintenance flexibility training programme of one session per week can help preserve the training gains (30). Similar to elastic (acute) ROM responses, plastic ROM changes or changes attributed to flexibility training can also be attributed to a variety of factors including neural, morphological, and psychological. Which of these factors is most predominant is still under debate?

Plastic Neural Adaptations

Is it possible that the neural inhibitory responses that occur with elastic or acute changes in ROM can persist or be semi-permanent with training or plastic changes? Just as with the neural adaptations associated with strength training, prolonged alterations in the nervous system can evolve with flexibility training. Blazeovich et al. (5) incorporated a three-week stretch training programme of the plantar flexors and found a reduction in tonic Ia (facilitatory) afferent feedback from muscle spindles (measured from the T-reflex). So, when the spindles were stretched, their firing frequency was less than before the flexibility training resulting in less reflex-induced contractions and a more relaxed muscle (disfacilitation). Perhaps, the decreased spindle activity was due to an increased compliance (reduced stiffness) of the passive elastic components. Thus, with a certain degree of stretch, the tissues around the spindles would accommodate the elongation and the spindle would not be stretched or elongated to the same extent resulting in a muted reflex response. However, in their study, the T-reflex reduction (spindle activity) did not parallel the reduction in passive stiffness during the stretch-training programme. Therefore, if the reduced spindle activity cannot be attributed to lower tissue compliance, then it is more likely that the flexibility training led to an intrinsic reduction in muscle spindle sensitivity. In this case, the stretch training had a direct effect on the activity of the nervous system (afferent input to the motoneurons). In the same Blazeovich study, they also reported increased

reciprocal inhibition in the soleus and gastrocnemius which would decrease the contractile force of the antagonists. Guissard and Duchateau had subjects stretch five days/week for six weeks (four stretch positions x five repetitions x 30 seconds each) and tested immediately after the training programme as well as after one month of detraining. Similar to the Blazevich et al. findings, they also found decreases in H- and T-reflexes (–36 percent and –14 percent respectively). However, they found that the reflex inhibition was present after the first ten stretching sessions but returned to baseline when tested after 30 days of detraining. However, passive muscle stiffness was apparent at every point from ten days of training to 30 days of detraining (15).

Plastic Morphological Adaptations

Stretch training studies in animal models have demonstrated sarcomerogenesis (increased number of sarcomeres in series) (8), but there is very limited or no evidence in humans. However, once again it should be pointed out that there are no longitudinal studies examining stretch-induced morphological changes over years of stretching. Again, with animal studies, stretch is a very potent signal for mechano growth factor (MGF: a variant of IGF-1), actin and myosin filament production, myosin isoform gene switching, protein turnover, hypertrophy as well as for adding sarcomeres in parallel and series (11, 12, 13, 14).

Traditionally PNF stretching effects were attributed primarily to neural inhibition (e.g., reciprocal inhibition, autogenic inhibition). However, there is a lack of experimental evidence for this emphasis on neural responses (6, 28). The effectiveness of PNF may be due to an increased tolerance to stretch (22) but there can also be morphological changes. For example, six weeks of PNF flexibility training reduced the passive and active stiffness of the Achilles tendon while also increasing the pennation angle of the gastrocnemius. However, there was no change in the passive resistive torque of the muscle (17). Even with 8–10-year old athletes, the flexibility trained athletes showed greater fascicle elongation when stretching with greater increases in joint angle and muscle-tendon unit displacement (27). Significant reductions in the myotendinous junction (muscle and tendon intersection) stiffness (e.g., 47 percent) during a passive static stretch have also been reported after stretch training (25) and this decreased passive stiffness can contribute to increased flexibility (15). On the contrary, a six week (five days/week) static stretch training programme improved hip flexion ROM but did not change muscle extensibility (4). The authors therefore attributed the improved ROM to increased stretch tolerance. Static stretch training (two days/week for 20 days) has been reported to reduce the tendon viscoelastic properties and but not the tendon stiffness (18) Static stretch training (seven days/week for six weeks) improved dorsiflexion ROM with a reduced muscle passive resistive torque but no change in tendon stiffness whereas ballistic stretch training exhibited the opposite effect with no change of muscle passive resistive torque but

decreased tendon stiffness (23). A review on chronic stretching changes reported that stretch training durations of 3–8 weeks (average study: 5.1 weeks) do not alter muscle or tendon properties although it can increase the extensibility and tolerance of the muscle to a greater tensile force (10). They also reported that these durations of stretching also have trivial effects on tendon properties. While their review supports the stretch tolerance theory, you have to ask yourself if most athletes or fitness enthusiasts only stretch for five weeks or less than two months. Many athletes will have stretched on a regular basis from adolescence to early adulthood and may continue on a less vigorous or consistent basis for many years hence. Thus, examining studies of only a maximum of eight weeks duration does not provide a full picture of possible chronic morphological changes.

It would be fantastic for the public if the studies found consistent results, but each study may use different types of stretching, durations, volumes, intensities and subject populations. The most prudent comment or recommendation might be to incorporate a variety of stretching styles (static, dynamic, and PNF) in order to ensure that morphological changes to the muscle and tendons is optimized by providing a variety of stressors to the system.

Psychological Adaptations

While the training-induced increase in joint ROM can be connected to neural and morphological alterations, there is also a strong case for an increase in stretch tolerance (21, 22). This is also known as the sensory theory, which indicates the musculotendinous unit can tolerate greater tensions without a change in tension for a given length. Sixty individuals stretched for 30 minutes; five times per week for six weeks with no change in hamstrings' extensibility. As there was an increase in the ROM, the authors suggest that the improvement must have been due to greater stretch tolerance (4).

Summary

While there is still conflict in the literature, generally static and PNF stretching tend to provide greater improvements with static ROM than dynamic and ballistic stretching. Flexibility training adaptations can persist for 3–8 weeks with either reduced (once a week) or minimal stretching and activity. This persistent flexibility adaptations may be partially ascribed to neural adaptations such as an intrinsic disfacilitation of spindle afferent discharge. Although animal stretching studies have shown an increase in sarcomeres in series, there are no similar human findings with flexibility training. However, there is evidence with human chronic flexibility training for alterations in muscle pennation angles, visco-elastic properties and stretch tolerance.

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